Title:

"Finite Element Modelling of a Timber Plate Using LS-DYNA: A Technical Project on Progressive Damage Analysis"

Prepared By: Imran Sarker Siam

Affiliations: Department of Building Engineering & Construction Management, Rajshahi University of Engineering & Technology, Bangladesh

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Project Overview

This project presents the finite element modelling (FEM) of a timber plate subjected to uniaxial tensile loading using LS-DYNA. The primary objective is to evaluate the structural response and damage behaviour of timber through the application of a damage-based material model (MAT_054).

The study was carried out as a technical engineering project to demonstrate practical competency in numerical modelling, mesh discretisation, boundary condition implementation, and simulation result interpretation. Particular emphasis was placed on capturing progressive damage evolution and post-peak softening behaviour, which are critical for understanding the mechanical performance of timber and composite materials.

1. Introduction:

The mechanical behaviour of structural materials under loading plays a critical role in predicting strength, stiffness, and failure mechanisms. In materials such as timber and timber-based composites, the response is inherently anisotropic and nonlinear, often characterised by progressive damage rather than sudden brittle failure. This makes the accurate prediction of their structural performance particularly challenging using conventional analytical methods.

With the advancement of computational mechanics, Finite Element Analysis (FEA) has become an essential tool for capturing complex material behaviour. In particular, explicit solvers such as LS-DYNA enable detailed simulation of damage initiation, stiffness degradation, and post-peak softening, which are critical for understanding failure mechanisms in composite-like materials.

In this study, a finite element model of a timber plate subjected to uniaxial tensile loading is developed to investigate its mechanical response. A damage-based material model (MAT_054) is employed to simulate progressive failure behaviour. The analysis focuses on extracting stress–time, displacement–time, and stress–strain responses to provide a comprehensive understanding of the structural behaviour.

2. Problem Statement:

The accurate prediction of damage and failure behaviour in timber structures remains a significant challenge due to their anisotropic nature and progressive failure characteristics. Unlike conventional isotropic materials, timber exhibits gradual stiffness degradation and post-peak softening, which cannot be effectively captured using simplified elastic or linear models. Traditional design approaches often neglect the nonlinear damage evolution that governs failure in such materials, leading to potential inaccuracies in estimating structural performance and safety. Moreover, the ability to simulate post-peak behaviour is essential for understanding residual strength and deformation capacity, especially in advanced timber and composite systems.

Although Finite Element Analysis provides a powerful framework for modelling such behaviour, the reliability of the results strongly depends on the selection of appropriate constitutive models capable of representing damage initiation and evolution.

Therefore, there is a need to evaluate the effectiveness of damage-based material models in capturing the complete mechanical response of timber under tensile loading conditions, including elastic behaviour, damage progression, and post-failure softening.

3. Objectives:

The main objectives of this project are:

- To develop a finite element model of a timber plate under uniaxial tensile loading using LS-DYNA
- To implement a damage-based material model (MAT_054) for simulating progressive failure behaviour
- To analyse the stress–time and displacement–time responses of the model
- To derive the stress–strain relationship and characterise material behaviour
- To investigate the initiation and evolution of damage and its effect on structural response
- To evaluate the capability of the numerical model in capturing post-peak softening and residual strength

4. Numerical Model:

A rectangular plate with a total length of 140 mm was modelled. The specimen includes 30 mm grip regions on both ends, resulting in an effective gauge length of 80 mm, which represents the deformable region used for strain calculations.

The geometry was discretised using a structured finite element mesh to ensure numerical stability and accurate stress distribution. Boundary conditions were applied such that one end of the plate was constrained, while the other end was subjected to displacement-controlled loading along the longitudinal direction.

5. Material Modelling:

The material behaviour was defined using **MAT_054**, a composite damage model available in LS-DYNA. This model is capable of simulating:

- Elastic behaviour in the initial loading stage
- Progressive damage initiation based on failure criteria
- Stiffness degradation after failure
- Residual load-carrying capacity

Such a formulation is suitable for representing composite or anisotropic materials where failure occurs gradually rather than instantaneously.

6. Loading and Analysis Procedure:

A displacement-controlled loading condition was applied to the model. Initially, the loading curve exhibited a delayed ramp, which resulted in a non-physical flat region in the response. This was corrected by modifying the loading definition to ensure that deformation begins at **time t = 0**.

The analysis was carried out over a defined simulation time, during which the stress and displacement histories were recorded. These time-dependent results were later processed to obtain the stress–strain relationship.

7. Mesh Discretisation and Element Size:

The rectangular plate was discretised using a structured finite element mesh with an approximate element size of 2 mm along both longitudinal and transverse directions. This resulted in a sufficiently refined mesh capable of capturing the stress distribution and deformation behaviour with adequate resolution.

The choice of a 2 mm element size was made based on the requirement to balance computational efficiency with numerical accuracy. For a plate of 140 mm length, this corresponds to approximately 70 elements along the loading direction, which is sufficient to resolve stress gradients and avoid artificial stiffness effects.

The adopted mesh size ensures:

- Accurate representation of stress localisation near the loaded region,
- Smooth variation in the displacement field,
- Proper capture of damage initiation and post-peak softening behaviour associated with MAT_054.

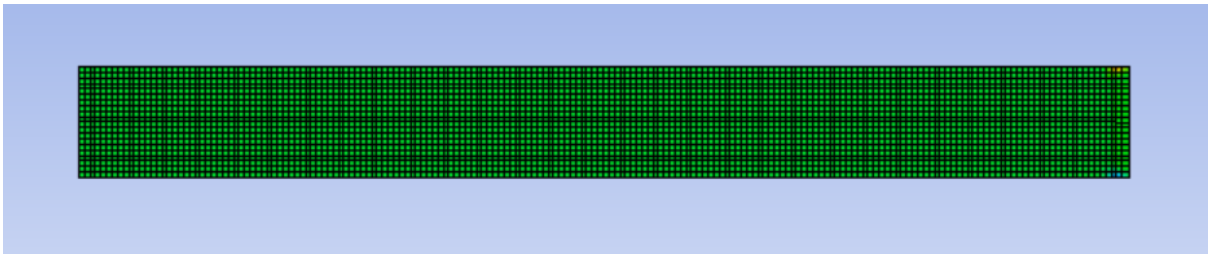

Figure 1: *Structured finite element mesh of the rectangular plate model with a uniform element size of approximately 2 mm, resulting in a refined discretisation along the 140 mm length.*

8. Results and Discussion:

8.1 Stress–Time Response:

The stress–time response, shown in Figure 2-3, initially exhibits a linear increase, corresponding to the elastic behaviour of the material. In this region, the stiffness remains constant, and the response is governed by the initial elastic properties defined in the material model.

As loading progresses, deviation from linearity is observed, indicating the initiation of damage within the material. This nonlinearity arises from the progressive degradation of stiffness governed by the damage evolution formulation in MAT_054. The peak stress of approximately 27 MPa represents the maximum load-carrying capacity of the timber plate.

Beyond this point, a sharp reduction in stress is observed, which is associated with the propagation of damage and loss of material integrity. Unlike brittle failure, the gradual decline in stress demonstrates the capability of the model to capture progressive failure mechanisms, where micro-level damage accumulates and leads to macroscopic softening.

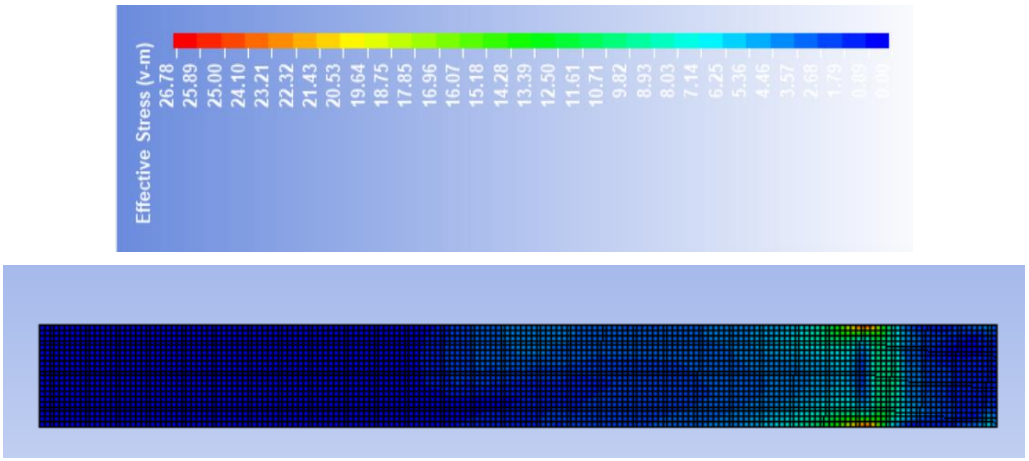

Figure 2: *Effective (von Mises) stress distribution in the plate obtained from the LS-DYNA simulation.*

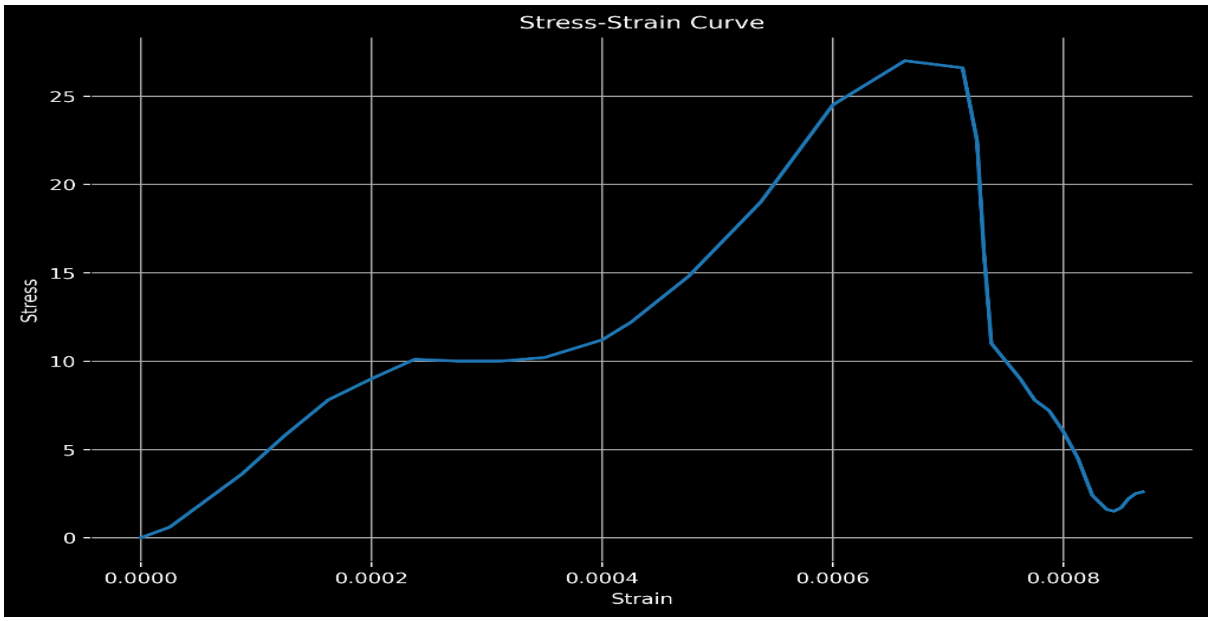

Figure 3: *Stress–time response of the plate obtained from the LS-DYNA simulation.*

8.2 Displacement–Time Response

The displacement–time response shows a smooth and continuous increase throughout the simulation, indicating stable numerical behaviour that is shown in Figure 4. The absence of oscillations confirms that the chosen mesh and time integration parameters ensure computational stability.

A noticeable reduction in slope occurs near the peak stress region, which corresponds to the onset of damage and reduction in effective stiffness. This behaviour reflects the transition from elastic to damage-dominated response.

Importantly, displacement continues to increase even after the stress begins to decrease. This indicates that the structure retains deformation capacity despite loss of strength, a characteristic feature of materials exhibiting strain-softening behaviour. This phenomenon is critical in understanding ductility and energy dissipation capacity.

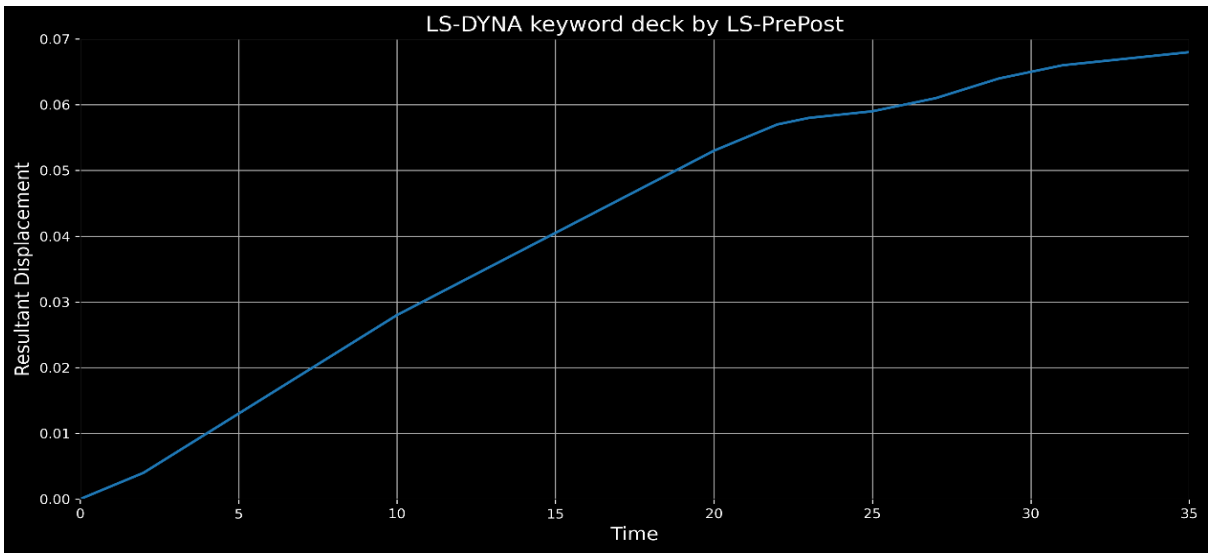

Figure 4: *Displacement–Time Response of the plate obtained from the LS-DYNA simulation.*

8.3 Stress–Strain Behaviour:

The stress–strain response shown in Figure 5 provides a comprehensive representation of the material behaviour. The initial linear portion corresponds to elastic deformation, where stress is directly proportional to strain.

The subsequent nonlinear region marks the initiation and evolution of damage, where stiffness progressively decreases due to internal material degradation. This behaviour is governed by the damage parameters in MAT_054, which reduce the effective stiffness as strain increases.

The peak stress of approximately 27 MPa represents the ultimate strength of the material. Beyond this point, the descending branch of the curve indicates strain-softening behaviour, where stress decreases with increasing strain.

This post-peak response is particularly significant, as it demonstrates the ability of the numerical model to capture progressive failure rather than abrupt fracture. The presence of a residual tail further indicates that the material retains limited load-carrying capacity even after significant damage, which can be attributed to partial stress redistribution within the structure.

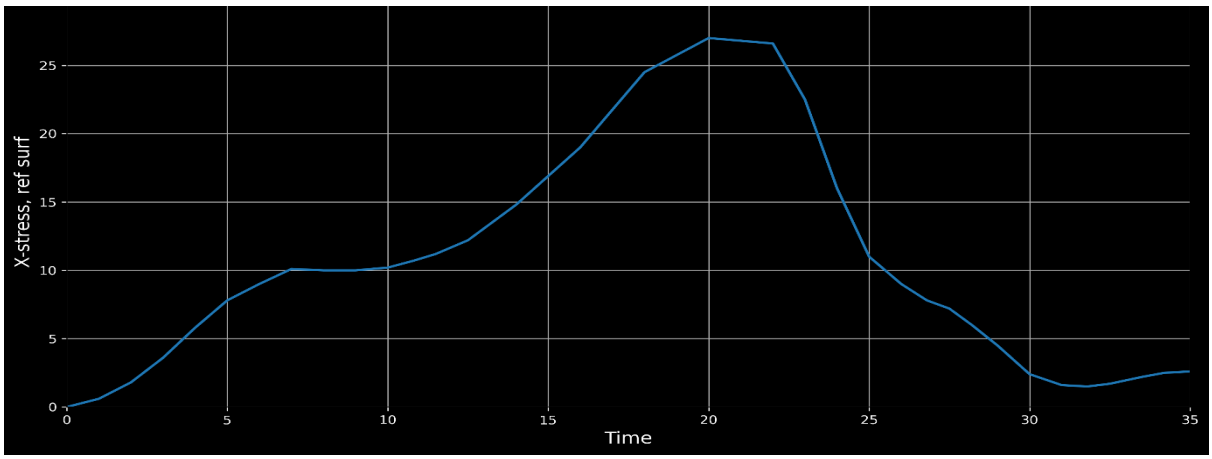

Figure 5: *Stress–Strain Behaviour of the plate obtained from the LS-DYNA simulation.*

9. Mechanical Interpretation:
The simulation demonstrates a progressive failure mechanism rather than brittle fracture. The key observations include:

- Gradual transition from elastic to nonlinear behaviour
- Peak strength followed by stiffness degradation
- Continued deformation after failure initiation
- Residual strength due to partial load redistribution

Such behaviour is typical of fibre-reinforced or layered materials, where damage initiates locally and propagates progressively.

10. Engineering Significance and Applications:
The outcomes of this project demonstrate the practical capability of finite element modelling in analysing the structural response and failure behaviour of timber-like materials. The ability to capture progressive damage and post-peak softening is particularly important for engineering applications where safety and reliability are critical.

The modelling approach adopted in this study can be applied in:

- Structural safety assessment – to evaluate load-carrying capacity and failure progression in timber elements
- Timber–composite systems – for analysing hybrid structures combining timber with fibre-reinforced materials
- FEM-based design validation – to support numerical verification of structural components before physical testing
- Damage modelling applications – for studying stiffness degradation and failure mechanisms in anisotropic materials

11. Conclusion
A finite element model of a rectangular timber plate was successfully developed and analysed using LS-DYNA with MAT_054. The simulation captured the full mechanical response, including elastic deformation, peak strength, and post-failure softening.
The corrected loading definition ensured physically meaningful results, and the conversion to a stress–strain curve provided a clear representation of material behaviour. The study confirms that the adopted modelling approach is suitable for analysing progressive damage in advanced material systems.

12. Skills Demonstrated

- Finite Element Modelling (FEM)
- LS-DYNA Simulation and Pre/Post Processing
- Mesh Discretisation and Numerical Stability Analysis
- Constitutive Damage Modelling for Anisotropic Materials
- Structural Behaviour and Failure Analysis
- Engineering Interpretation of Simulation Results